

Block-pass connectivity is supportable at 15°

RULING - The historical incidence proxy passes without using the 25° widening. Six of eight panels pass the frozen numerical rule; all eight are reported below.

All eight required combinations

City	Season window	View	Blocks	Passes	Block-passes	Median	P10	Blocks <8	LCC share	Sectors	Result
PHX	Primary	≤15°	2,870	5	14,087	5.0	5.0	2,870	1.000	4/4	FAIL
PHX	Primary	≤25°	2,870	13	35,062	12.0	11.0	0	1.000	4/4	PASS
PHX	Sensitivity	≤15°	2,870	15	39,887	14.0	14.0	0	1.000	4/4	PASS
PHX	Sensitivity	≤25°	2,870	38	101,500	35.0	34.0	0	1.000	4/4	PASS
LA	Primary	≤15°	4,286	25	95,544	22.0	21.0	0	1.000	4/4	PASS
LA	Primary	≤25°	4,286	43	170,528	40.0	38.0	0	1.000	4/4	PASS
LA	Sensitivity	≤15°	4,286	8	24,121	6.0	5.0	4,286	1.000	4/4	FAIL
LA	Sensitivity	≤25°	4,286	15	52,037	12.0	11.0	0	1.000	4/4	PASS

Frozen pass rule: ≥30 blocks, ≥3 sectors, largest-component share ≥0.80, and median ≥8 passes/block. LCC is the share of all eligible block-pass edges in the largest bipartite component.

The two failures are count-limited: Phoenix primary ≤15° has five eligible passes; Los Angeles sensitivity ≤15° has median 6.0 passes/block. Both pass after the required 25° sensitivity is displayed, but that widening is not needed for the overall ruling.

What this report establishes - and what it does not

Decision

Proceed with connectivity planning at the 15° near-nadir set. Do not invoke the single 25° widening for the ruling. The 25° panels remain incidence sensitivities and are not fully geometry-admissible until view azimuth is resolved.

Spatial structure

Every panel contains all four projected urban sectors and has an LCC share of 1.000. The maps show broad urban-area coverage rather than an airport, desert-fringe, or single-park island. The largest modal land-use class is Developed, Medium Intensity in every panel, accounting for 60.1%-64.8% of eligible edges.

Files to inspect

figures/d1b_map.png maps the eight panels; figures/d1b_hist.png shows their pass-count distributions; tables/d1b_connectivity.csv is the machine-readable eight-row result.

Method in one paragraph

The audit uses fixed 1 km local-UTM blocks whose centroids fall within each repaired 2020 Census urban-area boundary. Candidate 2019-2025 passes must be quality-and-exact-weather complete, fall in the local-solar season, and meet the pass-level p95 view-zenith set. An edge needs ≥60 clear, valid-view native 70 m cells after canonical MGRS partitioning and a cloudy-wins overlap rule. Modal 2024 Annual NLCD Collection 1.1 class supplies the descriptive land-use label.

Mandatory limitations

This is inherited Collection 2 quality/geometry evidence - not the new v6.2 Collection 3 study sample. The 60-cell edge is an optimistic necessary condition for later disjoint 30-tree and 30-background floors; Science TCC, fixed-background, canopy-span, and optical checks can only remove edges.

The inherited five-city census begins June 1. Phoenix primary and Los Angeles sensitivity are therefore June 1-July 10 lower bounds for their nominal May 15-July 10 windows. No quality-and-weather-complete Phoenix May 15-31 augmentation pass was available.

No ECOSTRESS LST value and no new v6.2 coefficient was opened. This free check does not authorize Gate A, sample selection, or thermal processing.

Land-use dominance detail

City	Window	View	Dominant modal Annual NLCD class	Eligible-edge share
PHX	Primary	≤15°	Developed, Medium Intensity	60.2%
PHX	Primary	≤25°	Developed, Medium Intensity	60.1%
PHX	Sensitivity	≤15°	Developed, Medium Intensity	60.3%
PHX	Sensitivity	≤25°	Developed, Medium Intensity	60.3%
LA	Primary	≤15°	Developed, Medium Intensity	64.7%
LA	Primary	≤25°	Developed, Medium Intensity	64.8%
LA	Sensitivity	≤15°	Developed, Medium Intensity	64.5%
LA	Sensitivity	≤25°	Developed, Medium Intensity	64.8%

Annual NLCD source: USGS Annual NLCD Collection 1.1 Land Cover 2024, 30 m, nearest-neighbor export. Exact city hashes are frozen in analysis_manifest.json.